Calcurion Maths Worksheets

A-Level Mathematics

Topic: Statistics – Sampling & Data

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Define a population and a sample. **[2]**

2. Explain the difference between a census and a sample survey. **[3]**

3. State one advantage and one disadvantage of taking a census. **[2]**

4. Describe simple random sampling. **[3]**

5. Describe systematic sampling. **[3]**

6. Describe stratified sampling. **[3]**

7. Describe quota sampling. **[3]**

8. A school has 600 students: 360 in Year 12 and 240 in Year 13. A stratified sample of 50 is required. How many from each year? **[3]**

9. A list of 2000 customers is numbered 0001 to 2000. Describe how to select a simple random sample of 40. **[3]**

10. Explain what is meant by a sampling frame. **[2]**

11. Give one advantage of stratified sampling over simple random sampling. **[2]**

12. Give one disadvantage of quota sampling. **[2]**

13. A researcher stands outside a supermarket and asks every 10th person who exits. Name this sampling method and state one limitation. **[3]**

14. Distinguish between qualitative and quantitative data. Give an example of each. **[3]**

15. Distinguish between discrete and continuous data. Give an example of each. **[3]**

Section B – Hard (Q16–25)

16. A college has 800 students in three faculties: Arts 200, Science 350, Business 250. A stratified sample of 80 is required. Calculate the number from each faculty. **[4]**

17. Explain why a sampling frame might not be complete and how this could bias results. **[4]**

18. A company wants to survey employees. There are 5 departments of different sizes. Recommend a sampling method and justify it. **[4]**

19. Describe how to carry out systematic sampling from a list of 500 names to obtain a sample of 25. **[4]**

20. Compare the advantages and disadvantages of simple random sampling and opportunity sampling. **[5]**

21. A questionnaire asks “Do you agree that the excellent new policy should be introduced?” Criticise this question. **[3]**

22. Explain the difference between primary and secondary data. Give one advantage of each. **[4]**

23. A sample of 100 people is taken from a town of 50 000. The sample mean height is 172 cm. Can this be used to estimate the population mean? Comment. **[3]**

24. Design a brief sampling plan to investigate the average time students at a sixth form spend on homework each week. **[5]**

25. Explain what is meant by bias in sampling and give two ways it can arise. **[4]**

End of Worksheet • *Show all working clearly.*